

Name:

Class:

Date:

GCSE TOPIC TEST

PHYSICS

H

Higher Tier

4.5 Forces

Materials

For this paper you must have:

- a ruler
- a scientific calculator
- the Physics Equation sheet

Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- In all calculations, show clearly how you work out your answer.

Information

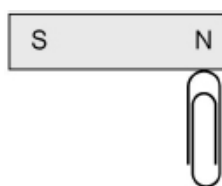
- The maximum mark for this paper is 40.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	

Figure 1 shows a paper clip being attracted to a permanent magnet.

Figure 1



- (a)** Write down the equation that links gravitational field strength (g), mass (m) and weight (W).

(1)

- (b)** The student added more paperclips to one end of the magnet.

The maximum number of paperclips the magnet could hold was 20.

Each paper clip had a mass of 1.0 g.

Gravitational field strength = 9.8 N/kg

Calculate the maximum force the magnet can exert.

Force = _____ N

(3)

[4 marks]

02.1

A student carried out an investigation to determine the spring constant of a spring.

Table 1 gives the data obtained by the student.

Table 1

Force in N	Extension in cm
0	0.0
2	3.5
4	8.0
6	12.5
8	16.0
10	20.0

The student measured the extension for five different forces rather than just measuring the extension for one force.

Suggest why.

[1 mark]

02.2

- (a) What is the equation that links extension, force and spring constant?

(1)

- (b) Calculate the spring constant of the spring that the student used.

Give your answer in newtons per metre.

Spring constant = _____ N/m

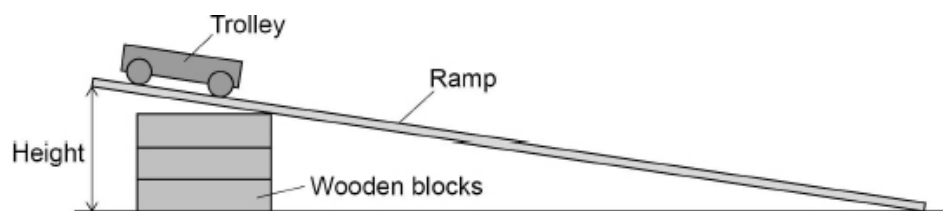
(4)

[5 marks]

A student investigated how the height of a ramp affects the acceleration of a trolley down the ramp.

Figure 2 shows some of the equipment used.

Figure 2

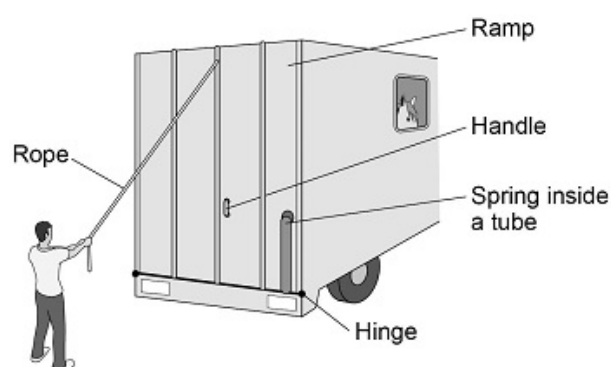


Plan an investigation to determine how the height of the ramp affects the acceleration of the trolley.

[6 marks]

Figure 3 shows the back of a lorry. The lorry is used to carry horses.

Figure 3



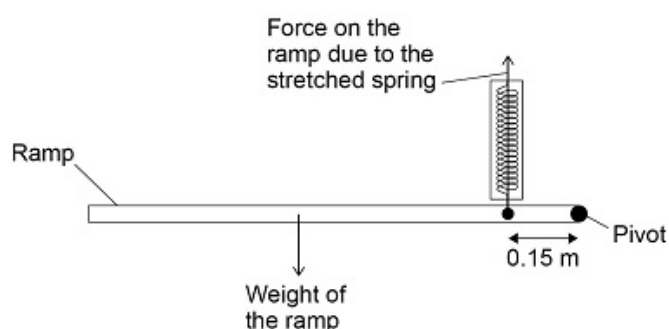
The ramp is lowered by pulling on the rope or by pulling on the handle.

The hinge acts as a pivot.

When the ramp is lowered, work is done to stretch a spring on the side of the ramp. Elastic potential energy is stored in the stretched spring.

Figure 4 shows the ramp part way down in a balanced horizontal position.

Figure 4



With the ramp horizontal:

The moment caused by the weight of the ramp = 924 Nm

The spring is stretched by 0.250 m

Calculate the elastic potential energy stored in the stretched spring.

Use data from **Figure 4**.

Elastic potential energy = J

[6 marks]

05

A person swims from a depth of 0.50 m to a depth of 1.70 m below the surface of the sea.

Density of the sea water = 1030 kg/m^3

Gravitational field strength = 9.8 N/kg

Calculate the increase in pressure on the swimmer.

Give the unit.

Use an equation from the Physics Equation Sheet.

Increase in pressure =

Unit

[4 marks]

06

An aeroplane lands at a speed of 80 m/s .

After landing, the aeroplane takes 28 s to decelerate to a speed of 10 m/s .

The mean resultant force on the aeroplane as it decelerates is $750\,000 \text{ N}$.

Calculate the mass of the aeroplane.

Mass = kg

[5 marks]

07

Figure 5 shows an ice skater, Skater **A**.

Figure 5



Skater **A** bumps into another skater, Skater **B**. Skater **B** is stationary.

The skaters move off together in a straight line.

Explain what happens to the velocity of each of the skaters.

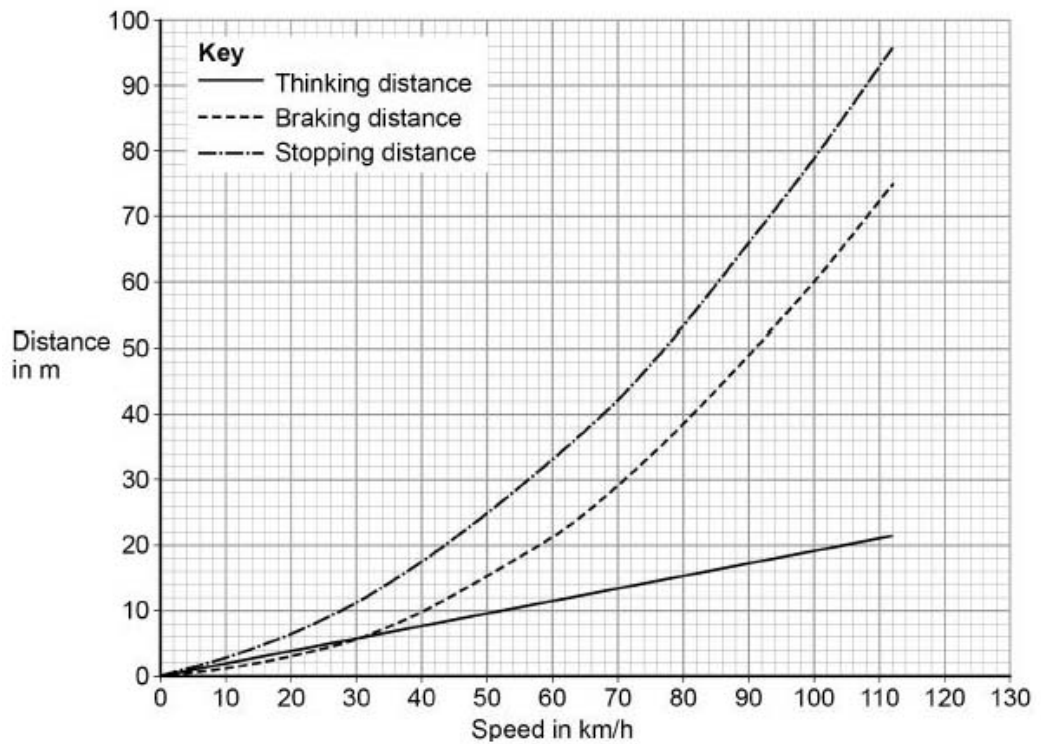
Use the idea of conservation of momentum.

[3 marks]

08

Figure 6 shows how the thinking distance, braking distance and stopping distance for a car vary with the speed of the car.

Figure 6



Describe the relationships shown in the graph above.

You should include factors that would affect the gradient of the lines.

[6 marks]